Comparación de los modelos - Pontia Logista

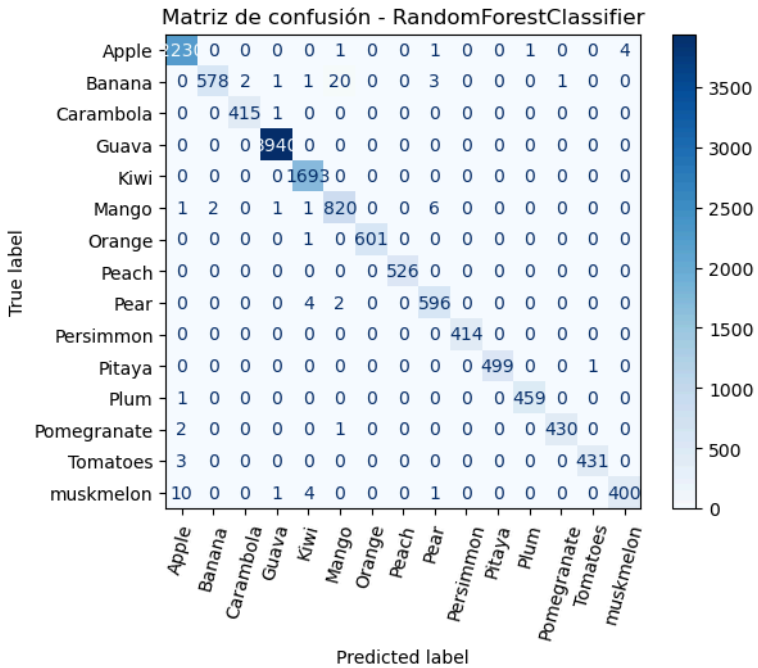
PONTIA LOGISTA	ML		DL	
	RandomForrestClassifier	KNN	MobilNetV2	CNN con TensorFlow
Accuracy	0.994472005669738	0.864209780297661	0.988163590431213	0.956410825252533
%	99.45%	86.42%	98.82%	95.64%

ML:

RandomForrestClassifier

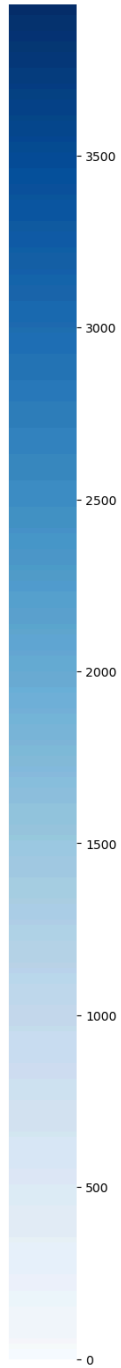
```
Accuracy RF: 0.9944720056697378
```

	precision	recall	f1-score	support
Apple	0.99	1.00	0.99	2237
Banana	1.00	0.95	0.97	606
Carambola	1.00	1.00	1.00	416
Guava	1.00	1.00	1.00	3940
Kiwi	0.99	1.00	1.00	1693
Mango	0.97	0.99	0.98	831
Orange	1.00	1.00	1.00	602
Peach	1.00	1.00	1.00	526
Pear	0.98	0.99	0.99	602
Persimmon	1.00	1.00	1.00	414
Pitaya	1.00	1.00	1.00	500
Plum	1.00	1.00	1.00	460
Pomegranate	1.00	0.99	1.00	433
Tomatoes	1.00	0.99	1.00	434
muskmelon	0.99	0.96	0.98	416
accuracy			0.99	14110
macro avg	0.99	0.99	0.99	14110
weighted avg	0.99	0.99	0.99	14110



Matriz de confusión - RandomForrestClassifier

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	2230	0	0	0	0	1	0	0	1	0	0	1	0	0	4
Banana	0	578	2	1	1	20	0	0	3	0	0	0	1	0	0
Carambola	0	0	415	1	0	0	0	0	0	0	0	0	0	0	0
Guava	0	0	0	3940	0	0	0	0	0	0	0	0	0	0	0
Kiwi	0	0	0	0	1693	0	0	0	0	0	0	0	0	0	0
Mango	1	2	0	1	1	820	0	0	6	0	0	0	0	0	0
Orange	0	0	0	0	1	0	601	0	0	0	0	0	0	0	0
Peach	0	0	0	0	0	0	0	526	0	0	0	0	0	0	0
Pear	0	0	0	0	4	2	0	0	596	0	0	0	0	0	0
Persimmon	0	0	0	0	0	0	0	0	0	414	0	0	0	0	0
Pitaya	0	0	0	0	0	0	0	0	0	0	499	0	0	1	0
Plum	1	0	0	0	0	0	0	0	0	0	0	459	0	0	0
Pomegranate	2	0	0	0	0	1	0	0	0	0	0	0	430	0	0
Tomatoes	3	0	0	0	0	0	0	0	0	0	0	0	0	431	0
muskmelon	10	0	0	1	4	0	0	0	1	0	0	0	0	0	400

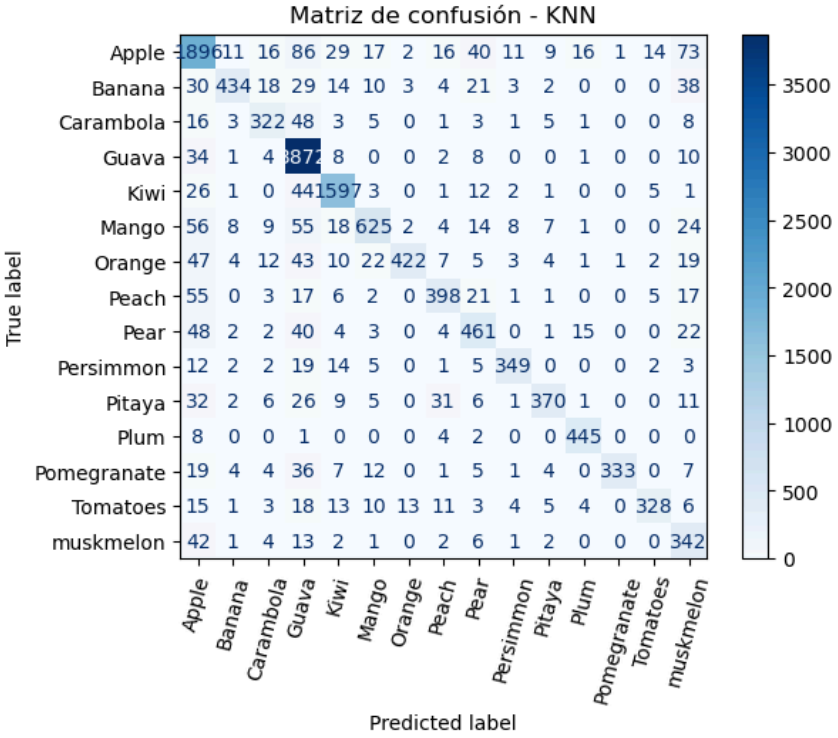


Matriz de confusión normalizada - RandomForrestClassifier

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Banana	0.00	0.95	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Carambola	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Guava	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Kiwi	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mango	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Orange	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Peach	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pear	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
Persimmon	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Pitaya	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00
Plum	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00
Pomegranate	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.00
Tomatoes	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00
muskmelon	0.02	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.96

KNN :

```
Accuracy: 0.8642097802976613

precision    recall  f1-score   support

       Apple       0.81      0.85      0.83      2237
      Banana       0.92      0.72      0.80       606
   Carambola       0.80      0.77      0.78       416
       Guava       0.89      0.98      0.93      3940
        Kiwi       0.92      0.94      0.93      1693
       Mango       0.87      0.75      0.81       831
      Orange       0.95      0.70      0.81       602
       Peach       0.82      0.76      0.79       526
        Pear       0.75      0.77      0.76       602
   Persimmon       0.91      0.84      0.87       414
      Pitaya       0.90      0.74      0.81       500
        Plum       0.92      0.97      0.94       460
 Pomegranate       0.99      0.77      0.87       433
    Tomatoes       0.92      0.76      0.83       434
   muskmelon       0.59      0.82      0.69       416

    accuracy                           0.86     14110
   macro avg       0.86      0.81      0.83     14110
weighted avg       0.87      0.86      0.86     14110
```



Matriz de confusión - KNN

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	1896	11	16	86	29	17	2	16	40	11	9	16	1	14	73
Banana	30	434	18	29	14	10	3	4	21	3	2	0	0	0	38
Carambola	16	3	322	48	3	5	0	1	3	1	5	1	0	0	8
Guava	34	1	4	3872	8	0	0	2	8	0	0	1	0	0	10
Kiwi	26	1	0	44	1597	3	0	1	12	2	1	0	0	5	1
Mango	56	8	9	55	18	625	2	4	14	8	7	1	0	0	24
Orange	47	4	12	43	10	22	422	7	5	3	4	1	1	2	19
Peach	55	0	3	17	6	2	0	398	21	1	1	0	0	5	17
Pear	48	2	2	40	4	3	0	4	461	0	1	15	0	0	22
Persimmon	12	2	2	19	14	5	0	1	5	349	0	0	0	2	3
Pitaya	32	2	6	26	9	5	0	31	6	1	370	1	0	0	11
Plum	8	0	0	1	0	0	0	4	2	0	0	445	0	0	0
Pomegranate	19	4	4	36	7	12	0	1	5	1	4	0	333	0	7
Tomatoes	15	1	3	18	13	10	13	11	3	4	5	4	0	328	6
muskmelon	42	1	4	13	2	1	0	2	6	1	2	0	0	0	342

Etiqueta real

Predicción

0 500 1000 1500 2000 2500 3000 3500

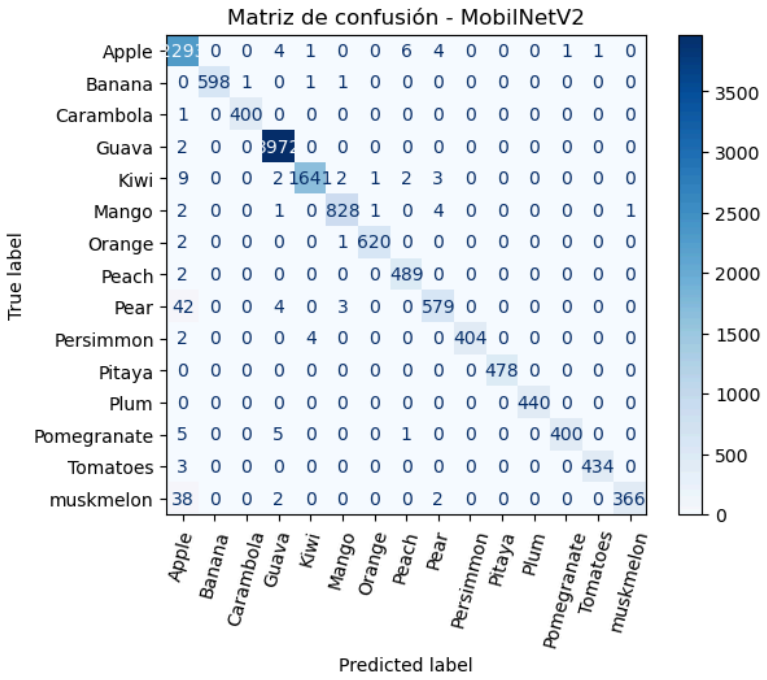
Matriz de confusión normalizada - KNN

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	0.85	0.00	0.01	0.04	0.01	0.01	0.00	0.01	0.02	0.00	0.00	0.01	0.00	0.01	0.03
Banana	0.05	0.72	0.03	0.05	0.02	0.02	0.00	0.01	0.03	0.00	0.00	0.00	0.00	0.00	0.06
Carambola	0.04	0.01	0.77	0.12	0.01	0.01	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.02
Guava	0.01	0.00	0.00	0.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Kiwi	0.02	0.00	0.00	0.03	0.94	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Mango	0.07	0.01	0.01	0.07	0.02	0.75	0.00	0.00	0.02	0.01	0.01	0.00	0.00	0.00	0.03
Orange	0.08	0.01	0.02	0.07	0.02	0.04	0.70	0.01	0.01	0.00	0.01	0.00	0.00	0.00	0.03
Peach	0.10	0.00	0.01	0.03	0.01	0.00	0.00	0.76	0.04	0.00	0.00	0.00	0.00	0.01	0.03
Pear	0.08	0.00	0.00	0.07	0.01	0.00	0.00	0.01	0.77	0.00	0.00	0.02	0.00	0.00	0.04
Persimmon	0.03	0.00	0.00	0.05	0.03	0.01	0.00	0.00	0.01	0.84	0.00	0.00	0.00	0.00	0.01
Pitaya	0.06	0.00	0.01	0.05	0.02	0.01	0.00	0.06	0.01	0.00	0.74	0.00	0.00	0.00	0.02
Plum	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.97	0.00	0.00	0.00
Pomegranate	0.04	0.01	0.01	0.08	0.02	0.03	0.00	0.00	0.01	0.00	0.01	0.00	0.77	0.00	0.02
Tomatoes	0.03	0.00	0.01	0.04	0.03	0.02	0.03	0.03	0.01	0.01	0.01	0.01	0.00	0.76	0.01
muskmelon	0.10	0.00	0.01	0.03	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.82

DL:

MobilNetV2

```
Validation accuracy: 0.9881635904312134

precision    recall  f1-score   support

       Apple       0.96      0.99      0.97      2310
      Banana       1.00      1.00      1.00       601
   Carambola       1.00      1.00      1.00       401
       Guava       1.00      1.00      1.00      3974
        Kiwi       1.00      0.99      0.99      1660
       Mango       0.99      0.99      0.99       837
      Orange       1.00      1.00      1.00       623
       Peach       0.98      1.00      0.99       491
        Pear       0.98      0.92      0.95       628
   Persimmon       1.00      0.99      0.99       410
      Pitaya       1.00      1.00      1.00       478
        Plum       1.00      1.00      1.00       440
 Pomegranate       1.00      0.97      0.99       411
    Tomatoes       1.00      0.99      1.00       437
   muskmelon       1.00      0.90      0.94       408

    accuracy                           0.99     14109
   macro avg       0.99      0.98      0.99     14109
weighted avg       0.99      0.99      0.99     14109
```



Matriz de confusión - MobilNetV2

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	2293	0	0	4	1	0	0	6	4	0	0	0	1	1	0
Banana	0	598	1	0	1	1	0	0	0	0	0	0	0	0	0
Carambola	1	0	400	0	0	0	0	0	0	0	0	0	0	0	0
Guava	2	0	0	3972	0	0	0	0	0	0	0	0	0	0	0
Kiwi	9	0	0	2	1641	2	1	2	3	0	0	0	0	0	0
Mango	2	0	0	1	0	828	1	0	4	0	0	0	0	0	1
Orange	2	0	0	0	0	1	620	0	0	0	0	0	0	0	0
Peach	2	0	0	0	0	0	0	489	0	0	0	0	0	0	0
Pear	42	0	0	4	0	3	0	0	579	0	0	0	0	0	0
Persimmon	2	0	0	0	4	0	0	0	0	404	0	0	0	0	0
Pitaya	0	0	0	0	0	0	0	0	0	0	478	0	0	0	0
Plum	0	0	0	0	0	0	0	0	0	0	0	440	0	0	0
Pomegranate	5	0	0	5	0	0	0	1	0	0	0	0	400	0	0
Tomatoes	3	0	0	0	0	0	0	0	0	0	0	0	0	434	0
muskmelon	38	0	0	2	0	0	0	0	2	0	0	0	0	0	366

Etiqueta real

Predicción

0 500 1000 1500 2000 2500 3000 3500

Matriz de confusión normalizada - MobilNetV2

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Banana	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Carambola	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Guava	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Kiwi	0.01	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mango	0.00	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Orange	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Peach	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pear	0.07	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.92	0.00	0.00	0.00	0.00	0.00	0.00
Persimmon	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.99	0.00	0.00	0.00	0.00	0.00
Pitaya	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00
Plum	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00
Pomegranate	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.97	0.00	0.00
Tomatoes	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.99	0.00
muskmelon	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.90

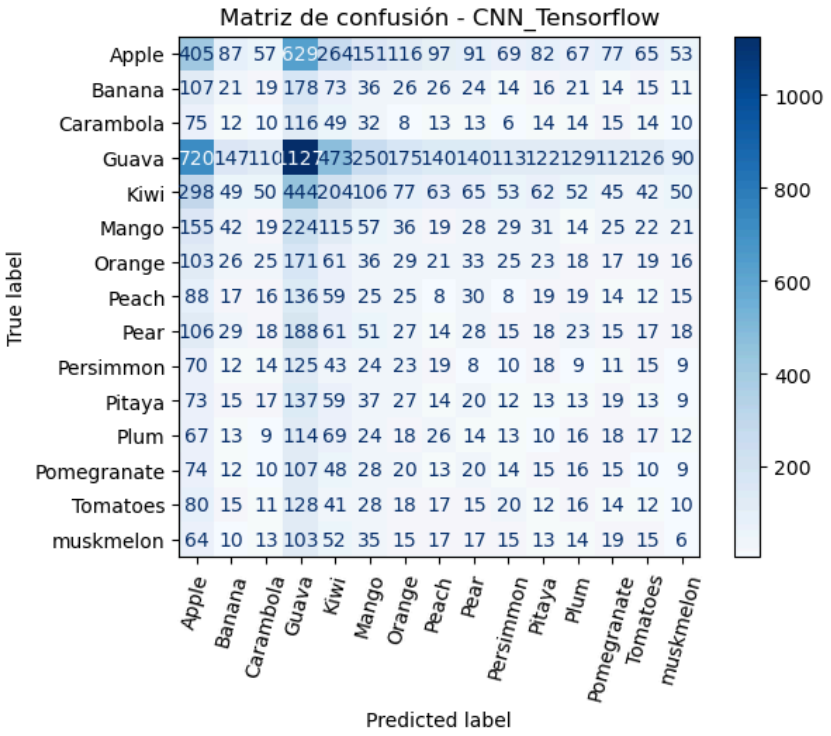
Etiqueta real

Predicción

1.0
0.8
0.6
0.4
0.2
0.0

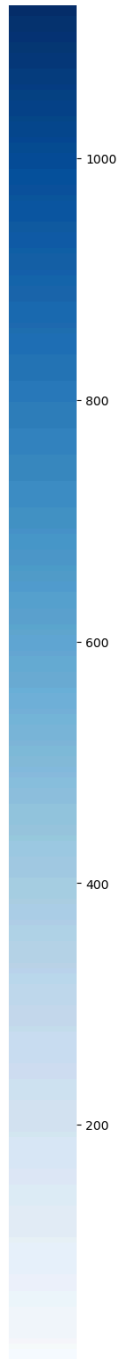
CNN_TensorFlow:

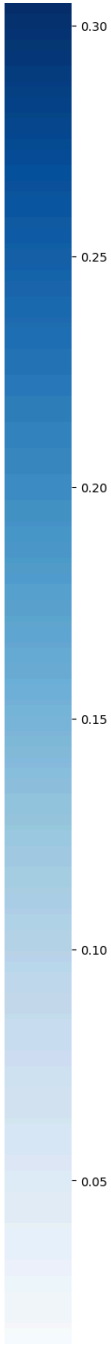
```
Accuracy: 0.956410825252533

precision    recall  f1-score   support

       Apple       0.16      0.18      0.17      2310
      Banana       0.04      0.03      0.04       601
   Carambola       0.03      0.02      0.03       401
       Guava       0.29      0.28      0.29      3974
        Kiwi       0.12      0.12      0.12      1660
       Mango       0.06      0.07      0.06       837
      Orange       0.05      0.05      0.05       623
       Peach       0.02      0.02      0.02       491
        Pear       0.05      0.04      0.05       628
   Persimmon       0.02      0.02      0.02       410
      Pitaya       0.03      0.03      0.03       478
        Plum       0.04      0.04      0.04       440
 Pomegranate       0.03      0.04      0.04       411
    Tomatoes       0.03      0.03      0.03       437
   muskmelon       0.02      0.01      0.02       408

    accuracy                           0.14     14109
   macro avg       0.07      0.07      0.07     14109
weighted avg       0.14      0.14      0.14     14109
```



Matriz de confusión - CNN_Tensorflow

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	405	87	57	629	264	151	116	97	91	69	82	67	77	65	53
Banana	107	21	19	178	73	36	26	26	24	14	16	21	14	15	11
Carambola	75	12	10	116	49	32	8	13	13	6	14	14	15	14	10
Guava	720	147	110	1127	473	250	175	140	140	113	122	129	112	126	90
Kiwi	298	49	50	444	204	106	77	63	65	53	62	52	45	42	50
Mango	155	42	19	224	115	57	36	19	28	29	31	14	25	22	21
Orange	103	26	25	171	61	36	29	21	33	25	23	18	17	19	16
Peach	88	17	16	136	59	25	25	8	30	8	19	19	14	12	15
Pear	106	29	18	188	61	51	27	14	28	15	18	23	15	17	18
Persimmon	70	12	14	125	43	24	23	19	8	10	18	9	11	15	9
Pitaya	73	15	17	137	59	37	27	14	20	12	13	13	19	13	9
Plum	67	13	9	114	69	24	18	26	14	13	10	16	18	17	12
Pomegranate	74	12	10	107	48	28	20	13	20	14	15	16	15	10	9
Tomatoes	80	15	11	128	41	28	18	17	15	20	12	16	14	12	10
muskmelon	64	10	13	103	52	35	15	17	17	15	13	14	19	15	6



Matriz de confusión normalizada - CNN_Tensorflow

Etiqueta real \ Predicción	Apple	Banana	Carambola	Guava	Kiwi	Mango	Orange	Peach	Pear	Persimmon	Pitaya	Plum	Pomegranate	Tomatoes	muskmelon
Apple	0.18	0.04	0.02	0.27	0.11	0.07	0.05	0.04	0.04	0.03	0.04	0.03	0.03	0.03	0.02
Banana	0.18	0.03	0.03	0.30	0.12	0.06	0.04	0.04	0.04	0.02	0.03	0.03	0.02	0.02	0.02
Carambola	0.19	0.03	0.02	0.29	0.12	0.08	0.02	0.03	0.03	0.01	0.03	0.03	0.04	0.03	0.02
Guava	0.18	0.04	0.03	0.28	0.12	0.06	0.04	0.04	0.04	0.03	0.03	0.03	0.03	0.03	0.02
Kiwi	0.18	0.03	0.03	0.27	0.12	0.06	0.05	0.04	0.04	0.03	0.04	0.03	0.03	0.03	0.03
Mango	0.19	0.05	0.02	0.27	0.14	0.07	0.04	0.02	0.03	0.03	0.04	0.02	0.03	0.03	0.03
Orange	0.17	0.04	0.04	0.27	0.10	0.06	0.05	0.03	0.05	0.04	0.04	0.03	0.03	0.03	0.03
Peach	0.18	0.03	0.03	0.28	0.12	0.05	0.05	0.02	0.06	0.02	0.04	0.04	0.03	0.02	0.03
Pear	0.17	0.05	0.03	0.30	0.10	0.08	0.04	0.02	0.04	0.02	0.03	0.04	0.02	0.03	0.03
Persimmon	0.17	0.03	0.03	0.30	0.10	0.06	0.06	0.05	0.02	0.02	0.04	0.02	0.03	0.04	0.02
Pitaya	0.15	0.03	0.04	0.29	0.12	0.08	0.06	0.03	0.04	0.03	0.03	0.03	0.04	0.03	0.02
Plum	0.15	0.03	0.02	0.26	0.16	0.05	0.04	0.06	0.03	0.03	0.02	0.04	0.04	0.04	0.03
Pomegranate	0.18	0.03	0.02	0.26	0.12	0.07	0.05	0.03	0.05	0.03	0.04	0.04	0.04	0.02	0.02
Tomatoes	0.18	0.03	0.03	0.29	0.09	0.06	0.04	0.04	0.03	0.05	0.03	0.04	0.03	0.03	0.02
muskmelon	0.16	0.02	0.03	0.25	0.13	0.09	0.04	0.04	0.04	0.04	0.03	0.03	0.05	0.04	0.01